Preprocessing Pipeline

CA Assignment 2 · Clustering Algorithms

Embedding model: `sentence-transformers/all-MiniLM-L12-v2`

Overview

The preprocessing pipeline transforms raw natural language sentences into a compact, geometrically well-structured numerical representation suitable for Agglomerative Hierarchical Clustering (AHC). The pipeline consists of four sequential stages:

1. Data loading and text cleaning
2. Sentence embedding using a pre-trained transformer model
3. L2 normalisation of the raw embedding vectors
4. Dimensionality reduction via Principal Component Analysis (PCA)

Each stage is motivated by theoretical considerations specific to text clustering and the intrinsic properties of the chosen embedding model.

1. Data Loading and Text Cleaning

The dataset is stored in a tab-separated file with two columns: an integer identifier (`ID`) and a raw sentence string (`Sentence`). The file is read line by line using a custom parser that identifies record boundaries by detecting lines whose first field is a positive integer followed by a tab character. Any line that does not match this pattern is treated as a continuation of the previous sentence, with its content appended and separated by a space. This approach correctly handles sentences that span multiple physical lines due to embedded newline characters in the source file, ensuring all 1,760 instances are loaded and each receives a cluster label. The resulting DataFrame has shape (1,760, 2), with the `Sentence` column containing the raw text strings.

1.1 Cleaning Strategy

A deliberate **light-touch cleaning strategy** is adopted, comprising only:

- Removal of escaped quotation marks (`\"` and `\'`).
- Collapsing of repeated whitespace characters (spaces, tabs) into a single space.
- Collapsing of actual newline characters into a single space.
- Stripping of leading and trailing whitespace.
- Replacement of empty strings with the placeholder token `'empty'` to preserve exact row alignment between the feature matrix and `label.txt`.

Heavy-handed normalisation operations — such as lowercasing, punctuation stripping, stopword removal, or stemming — are **intentionally excluded**. This decision is theoretically motivated: MiniLM is a transformer trained on natural, cased, punctuated English text using a contrastive sentence-pair objective. Its internal self-attention mechanism is sensitive to capitalisation (e.g., proper nouns such as CORIOLANUS carry identity and thematic meaning), punctuation (which marks prosodic and syntactic boundaries), and function words (which contribute to syntactic style and register). Removing any of these features would push sentences out of the model's training distribution, degrading embedding quality and consequently the discriminability of clusters.

Influence on clustering: Preserving natural sentence text ensures the embedding model operates within its training distribution, producing high-quality semantic representations. Embeddings derived from natural text exhibit greater intra-domain cohesion and inter-domain separation — the necessary geometric precondition for cluster recovery.

2. Sentence Embedding with MiniLM-L12-v2

Sentences are encoded into dense numerical vectors using the pre-trained model **`sentence-transformers/all-MiniLM-L12-v2`**, a 12-layer, 384-dimensional sentence transformer from the Sentence-BERT family (Reimers & Gurevych, 2019). The model was trained on over one billion sentence pairs using a Multiple Negative Ranking (MNR) loss — a contrastive objective that optimises the embedding space so that semantically similar sentences map to geometrically proximate points on the unit hypersphere in $\mathbb{R}^{384}$, while dissimilar sentences are pushed apart.

2.1 Why This Model?

- **Semantic richness.** Unlike bag-of-words representations (TF-IDF; Salton & McGill, 1983) or static word embeddings (Word2Vec, Mikolov et al., 2013; GloVe, Pennington et al., 2014), MiniLM produces contextual sentence-level embeddings that capture polysemy, syntactic structure, and discourse-level meaning. This is critical for a dataset spanning vastly different linguistic registers — from Shakespearean Early Modern English to contemporary informal prose — where the model must distinguish not only topic but also vocabulary era, syntactic style, and discourse register in order to produce an embedding space where thematically distinct groups are geometrically separable. It is important to note that the number of human-identifiable *source domains* in a dataset does not necessarily equal the number of geometric clusters the embedding model will produce. Two domains that are linguistically similar — such as restaurant reviews and news articles, which both employ contemporary vocabulary, modern syntax, and prose sentence structure — may be embedded closer to each other than to a third domain (e.g., Shakespearean dialogue), and correctly merge into one cluster under silhouette-optimal analysis. This is a property of the embedding geometry, not a failure of the model: the optimal $K = 3$ reflects three *geometrically* distinct regions of the embedding space, which need not correspond one-to-one with the number of original source corpora.

- **No cloud API dependency.** The model runs fully locally via the `sentence-transformers` library and Hugging Face's model hub, satisfying the assignment constraint that no cloud subscription is required.

- **Computational efficiency.** With 33 million parameters and a 384-dimensional output, MiniLM provides a strong balance between embedding quality and inference speed, making it practical for a 1,760-sentence dataset.

- **Cosine-similarity training objective.** The MNR loss explicitly trains the model to make cosine similarity between sentence embeddings proportional to semantic similarity. This makes the embedding space intrinsically well-suited to distance-based clustering algorithms.

- **Literature support.** MiniLM is built upon the BERT transformer architecture (Devlin et al., 2019) and trained using the Sentence-BERT framework (Reimers & Gurevych, 2019). Petukhova et al. (2024) demonstrate that BERT-family sentence transformers represent the best open-source option for text clustering, outperforming TF-IDF baselines and performing comparably to much larger open-source LLMs such as Falcon-7b and LLaMA-2.

2.2 Encoding Configuration

Sentences are encoded in batches of 64 with `normalize_embeddings=False`. Normalisation is deliberately disabled at this stage because it is applied explicitly in the subsequent pipeline step, maintaining clear separation of concerns. The output is a `float32` matrix of shape (1760, 384).

> **Influence on clustering:** The quality of the embedding is the single most important factor in text clustering performance. Sentences from the same domain (e.g., all Shakespearean dialogue) are embedded in geometrically proximate regions of $\mathbb{R}^{384}$, providing the raw signal from which clusters are recovered. No downstream processing step can recover structure that the embedding fails to encode.

3. L2 Normalisation

Each raw embedding vector $\mathbf{x} \in \mathbb{R}^{384}$ is L2-normalised:

$$\hat{\mathbf{x}} = \frac{\mathbf{x}}{\|\mathbf{x}\|_2} \qquad \text{such that} \qquad \|\hat{\mathbf{x}}\|_2 = 1$$

After normalisation, all vectors lie on the 383-dimensional unit hypersphere $\mathcal{S}^{383} \subset \mathbb{R}^{384}$.

3.1 Why Normalise Before PCA?

Without normalisation, raw MiniLM embeddings exhibit variation in both *direction* (semantic content) and *magnitude* (loosely correlated with sentence length and vocabulary richness). When PCA is applied to un-normalised vectors, it decomposes a mixture of directional and magnitude variance. Magnitude differences between sentences carry little semantic information relevant to thematic clustering. By placing all vectors on the unit sphere before PCA, we guarantee that PCA decomposes purely **angular (directional, semantic) variance** — the type most relevant for identifying thematic clusters.

A critical geometric consequence is that, for any two unit vectors $\hat{\mathbf{a}}, \hat{\mathbf{b}} \in \mathcal{S}^{383}$:

$$d_{\text{Euclidean}}(\hat{\mathbf{a}}, \hat{\mathbf{b}})^2 = \|\hat{\mathbf{a}} - \hat{\mathbf{b}}\|_2^2 = 2 - 2\langle\hat{\mathbf{a}}, \hat{\mathbf{b}}\rangle = 2(1 - \cos(\hat{\mathbf{a}}, \hat{\mathbf{b}}))$$

Euclidean distance is therefore a **monotone function of cosine distance** on the unit sphere. This means Ward-linkage AHC — defined in Euclidean space — implicitly optimises cluster separation in terms of cosine similarity, which is precisely the metric that MiniLM's training objective optimised.

> **Influence on clustering:** L2 normalisation before PCA ensures that principal components capture thematic (angular) variation rather than length-driven magnitude differences. The Euclidean–cosine equivalence on the unit sphere makes Ward-linkage AHC with Euclidean distance fully consistent with the semantic geometry of the embedding space.

4. Principal Component Analysis (PCA)

PCA (Pearson, 1901; Jolliffe, 2002) is applied to the L2-normalised matrix $\hat{X} \in \mathbb{R}^{1760\times384}$, projecting each sentence vector into the subspace spanned by the top 100 principal components:

$$X_{\text{pca}} = \hat{X}\,W, \qquad W \in \mathbb{R}^{384\times100}$$

where the columns of W are the eigenvectors of the sample covariance matrix $\hat{X}^\top\hat{X}$, ordered by decreasing eigenvalue. The 100 retained components collectively explain **74.55%** of the total variance in the normalised embedding matrix.

4.1 Why Apply PCA?

- **Curse of dimensionality.** In high-dimensional spaces, pairwise Euclidean distances become increasingly uniform (Beyer et al., 1999). Formally, for N random points in $\mathbb{R}^d$, the ratio $(d_{\max} - d_{\min})/d_{\min} \to 0$ as $d \to \infty$, meaning all points appear equidistant. This degrades any distance-based clustering algorithm. Reducing from 384 to 100 dimensions substantially mitigates this effect.

- **Noise suppression.** Components PC_{101} through PC_{384} capture noise and sentence-level idiosyncrasies rather than shared semantic structure. Discarding them acts as a low-rank denoising filter, retaining only the dominant directions of semantic variation useful for clustering.

- **Computational efficiency.** AHC computes an $O(N^2)$ pairwise distance matrix. Reducing feature dimensionality from 384 to 100 reduces memory and compute requirements substantially without sacrificing cluster quality.

- **Literature precedent.** Petukhova et al. (2024) use PCA as the sole dimensionality reduction step before applying clustering algorithms to BERT-family embeddings, validating this approach in the peer-reviewed literature.

4.2 Why 100 Components?

The choice of 100 components is motivated by the need to retain sufficient variance to preserve the fine-grained geometric separation between the small Wikipedia sub-corpus (90 documents) and the larger modern English cluster (880 documents). At 50 components, the cumulative explained variance is approximately 56.6%, and the Wikipedia signal — which resides in lower-ranked principal components that encode formal encyclopaedic register — is discarded, causing the silhouette-optimal K to reduce from 3 to 2. At 100 components, 74.55% of total variance is retained, preserving this fine-grained distinction and correctly recovering a three-cluster partition. Retaining too many components (e.g., 200+) reintroduces the curse of dimensionality; 100 strikes the correct balance between discriminative power and geometric stability.

4.3 The PCA Output Is NOT Re-Normalised — By Design

A critical design decision is that X_{pca} is used **directly**, without a second round of L2 normalisation. This is theoretically motivated.

PCA is a variance-preserving projection: PC_1 has the largest eigenvalue and the largest variance across the dataset; PC_2 the second largest; and so on. In the resulting 100-dimensional space, Euclidean distance naturally weights the earlier components more heavily, because they contribute more to the squared distance:

$$d^2(\mathbf{a}, \mathbf{b}) \;=\; \sum_{k=1}^{100}(a_k - b_k)^2$$

Since the earlier principal components capture the most semantically significant axes of variation (e.g., PC_1 encodes the dominant contrast between Shakespearean and modern English), this automatic variance weighting provides useful inductive bias for cluster discovery.

If X_{pca} were re-normalised, all 100 dimensions would be forced to contribute *equally* to distances, erasing this variance structure. Empirical validation confirmed this: re-normalising after PCA caused the optimal K to collapse from 3 to 2, merging the Wikipedia cluster into the larger modern English cluster, because the distinguishing features of the small Wikipedia sub-corpus reside in lower principal components that were amplified to equal weight by re-normalisation, reducing overall discriminability.

Influence on clustering: PCA's variance-weighted geometry causes Ward-linkage AHC to weight the most semantically informative dimensions most heavily. This enables the algorithm to recover fine-grained clusters — such as separating the small Wikipedia sub-corpus (90 documents) from the larger reviews/news cluster (880 documents) — that collapse to a trivial two-cluster partition in a uniformly-weighted re-normalised space.

5. Pipeline Summary

Stage	Operation	Input	Output	Key Purpose
1	Data loading & cleaning	Raw text file	1,760 sentences	Quality & row alignment
2	MiniLM-L12-v2 encoding	1,760 sentences	(1760, 384)	Semantic representation
3	L2 normalisation	(1760, 384)	(1760, 384) unit vectors	Angular variance for PCA
4	PCA (100 components)	(1760, 384)	(1760, 100) variance-weighted	Dimensionality reduction

Table 1: Summary of the four-stage preprocessing pipeline.

The final output is a `float32` matrix of shape (1760, 100) in which each row is the variance-weighted PCA projection of the corresponding sentence's L2-normalised MiniLM embedding. This matrix is passed directly to the AHC clustering algorithm.

6. Holistic Discussion: Influence on Clustering Performance

The preprocessing choices form a coherent, theoretically motivated pipeline specifically designed for sentence-level text clustering with a transformer embedding model. The key stage-to-stage interactions are:

- **Cleaning → Embedding quality.** Preserving natural text keeps sentence representations within MiniLM's training distribution. Sentences from semantically distinct domains (Shakespeare, reviews, Wikipedia) produce embeddings in clearly separated regions of $\mathbb{R}^{384}$.
- **L2 normalisation → PCA quality.** Normalising before PCA ensures that principal components reflect semantic (angular) variation rather than sentence-length variation, maximising discriminative information per component.
- **PCA → AHC quality.** Reducing to 100 variance-weighted dimensions eliminates noise, mitigates the curse of dimensionality, and concentrates discriminative semantic information in the first few components, which Ward-linkage AHC naturally weights more heavily.
- **Variance weighting → Cluster granularity.** Not re-normalising after PCA preserves the hierarchical weighting of semantic axes, enabling AHC to recover a three-cluster solution reflecting genuine thematic structure rather than collapsing to a trivial two-cluster partition.

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